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Paleomagnetism of the Beemerville (New Jersey) Alkaline Complex

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ABSTRACT

Stable remanent magnetism in the Late Ordovician Beemerville (New Jersey) alkaline complex (435 m.y.) is carried by secondary hematite resulting from supergene weathering prior to deposition of the Lower Silurian Shawangunk Conglomerate. The mean vector from four stable sites (of eight sampled) is D = 163°, I = 22° (κ = 23, α_{95} = 20°) and gives a virtual geomagnetic pole at 35°N., 126°E., δp = 9°, δm = 18°.

INTRODUCTION

The Beemerville alkaline complex, comprising nepheline syenite, tinguaite dikes, and ouachitite breccia pipes (Wolff, 1908; Aurosseau and Washington, 1922) is located in the northwest corner of the Franklin Furnace quadrangle in Sussex County, New Jersey (lat 41°14′ N., long 74°42′ W.; Fig. 1). The main body of nepheline syenite, about 2 km long and 300 to 500 m wide, forms a conspicuous northeast-trending ridge along the southeast flank of Kittatinny Mountain. The largest main ouachitite breccia plug (one of four), known locally as Rutan's hill or Clark's volcano, is 500 m in diameter, and is located near the northeastern tip of the syenite body. These occurrences have been the subject of petrologic investigations for many years (see, for example, Wilkerson, 1946; Davidson, 1948; Parker, 1948).

The syenite crops out along or near the contact between the Martinsburg Shale (Ordovician) and the Shawangunk Conglomerate (Lower Silurian; Kummel, 1940). It is intrusive into the former and apparently overlain by the latter, with a slight angular unconformity between the two (Kummel, 1940). Both formations now strike northeast and dip 20° to 30° to the northwest.

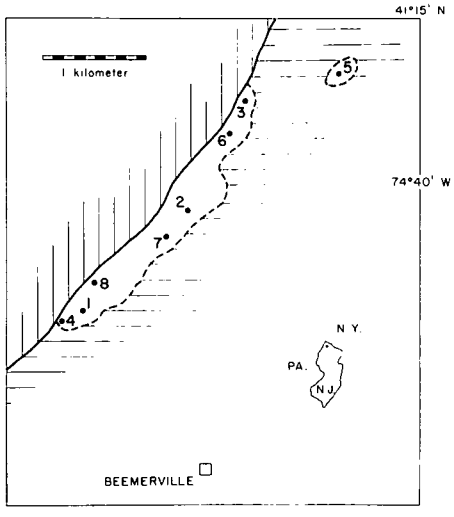


Figure 1. Location map and geologic sketch of the Beemerville nepheline syenite complex. Vertical and horizontal shadings mark Shawangunk (Lower Silurian) and Martinsburg (Upper Ordovician) Formations, respectively. Igneous complex is unshaded; numbered points are paleomagnetic sampling sites.

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The stratigraphic relations give an approximate Late Ordovician age for the complex. In agreement with this, Zartman and others (1967) have reported an age of 435 ± 20 m.y., based on K-Ar, Rb-Sr, and lead-alpha age determinations. This paper reports the result of a paleomagnetic study of the complex.

FIELD AND LABORATORY METHOD

At each of eight separate sites (Fig. 1), an average of five cores was drilled, or hand samples collected, oriented by means of a sun compass (Larochelle, 1961). At least two core specimens, 2.2 cm long, cut from each sample, were used in the magnetic measurements.

Orientation and intensity of natural remanent magnetization (NRM) were measured on a Princeton Applied Research SM-1 spinner magnetometer. Alternating field (AF) demagnetization was performed on standard equipment described by McElhinny (1963). Following reconnaissance, AF demagnetization of selected specimens up to 700 oe, all remaining specimens were demagnetized to 300 oe. The accuracy of individual vectors, because of orientation and measurement errors, is estimated to be ±5°. To correct for tectonic rotation all vectors have been rotated 25° E. about an axis striking N. 40° E., this being the average attitude of the Martinsburg and Shawangunk Formations.

RESULTS

The magnetic results obtained are presented in Table 1. With the exception of site 1, the NRM results from all sites are hopelessly scattered—the result from site 1 (D = 36°, I = +70°) is almost identical to the present local field (D = 33°, I = +69°) at Beemerville.

At AF demagnetization (see the representative curves in Fig. 2), a significant grouping of directions, quite distinct from the present Earth's field, was obtained in only four of the eight sites (Table 2), the mean of these (D = 163°, I = 22°, κ = 23, α_{95} = 20°) giving a pole at 35° N., 126° E., δp = 9°, δm = 18°.

Examination of nine polished thin sections from representative core specimens reveals a considerable variation in opaque mineral content (trace to 2 percent) and composition, even in specimens from the same site.

Pyrite is ubiquitous, sometimes co-

TABLE 1. SUMMARY OF MAGNETIC MEASUREMENTS*

Site	No. of samples	D_n	I_n	κ	$\bar{J}_n$ (emu/cc)	$\bar{X}$ (emu/cc oe)	J_n/X	D_{300}	I_{300}	κ	α_{95}	$(J_{300}/X)\times100$
1	5	36	70	8	2.2×10^{-3}	1.1×10^{-3}	2	40	78	1	--	0.45
2	5	104	36	4	1.4×10^{-3}	1.8×10^{-3}	0.8	143	30	6	35	2.0
3	5	165	15	2	1.1×10^{-4}	3.6×10^{-5}	3	170	4	83	8	50.0
4	3	309	49	1	5.1×10^{-5}	1.3×10^{-5}	4	17	14	1	--	4.0
5	6	129	42	3	2.0×10^{-4}	4.1×10^{-4}	0.5	167	24	7	28	6.0
6	4	297	76	4	2.5×10^{-4}	7.0×10^{-5}	3.5	172	26	16	24	17.0
7	6	149	55	2	2.1×10^{-3}	8.2×10^{-3}	0.25	122	-5	3	49	0.3
8	6	31	38	4	5.3×10^{-4}	3.2×10^{-3}	0.15	93	73	3	44	0.7

Mean	No. of sites	D_{300}	I_{300}	κ	α_{95}	Pole Lat	Pole Long	κ	α_{95}	δp	δm
2, 3, 5, 6	4	163	22	23	20	35° N.	126° E.	31	17	9	18

* *D = declination; I = inclination, with subscripts n and 300 referring to initial NRM, and after 300 oe AF demagnetization; $\bar{J}_n$ = average intensity of NRM; $\bar{X}$ = average magnetic susceptibility; κ and α_{95} = Fischer's precision parameter and circle of confidence, both to nearest whole number; δp and δm = semiaxes of polar error ellipse.*

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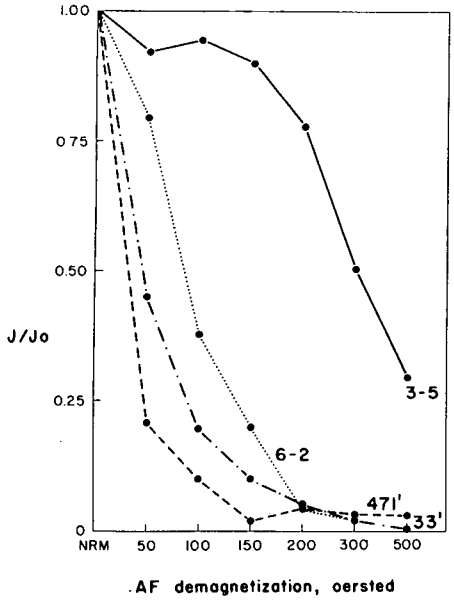


Figure 2. Normalized AF demagnetization curves, contrasting NRM coercivity spectra of surface samples (3-5, and 6-2) with fresh chips from deep boreholes (33 and 471 ft).

existing with pyrrhotite, both invariably showing variable degrees of alteration to fine-grained cryptocrystalline hematite. In some samples, hematite has formed at the expense of a pre-existing ferromagnesian mineral.

Magnetite is present as coarse (200 μ) crystals in three of the nine samples studied, with rare exsolved ilmenite lamellae in one instance. While the coexisting sulfide is always variably altered to hematite, the magnetite, when present, appears quite fresh.

Through the courtesy of Larry McKague (then at Rutgers University), we obtained nine samples from varying depths (to a maximum of 471 ft) in three boreholes drilled in the Beemerville syenite. Observations of the opaque mineralogy in these chips compare with the surface samples as follows:

1. The sulfide is almost invariably fresh and unaltered; in only one chip from a depth of 33 ft was hematite alteration conspicuous. In one sample from a depth of 358 ft, specular hematite appears to have formed from pre-existing carbonate mineral.

2. Magnetite was observed in every borehole core specimen, although it is present as a substantial opaque constituent only in four (of the nine) specimens taken from depths of from 60 to 220 ft. Pyrrhotite is the predominant sulfide in specimens from 324- and 471-ft depths.

The restriction to surface samples of the pervasive alteration of sulfide and ferromagnesian minerals (and possibly magnetite) to hematite suggests that the hematite is the product of supergene weathering and oxidation. This fact is important in the interpretation of the paleomagnetic results.

TABLE 2. LOWER PALEOZOIC POLE POSITIONS FROM NORTH AMERICA

No.*	Pole	
	Lat	Long
1	35° N.	126° E.
2	36° N.	114° E.
3	20° N.	153° E.
4	32° N.	102° E.
5	27° N.	137° E.
6	29° N.	119° E.
7	43° N.	127° E.

** 1 = Beemerville alkaline complex, this study; 2 = Trenton limestone (Ordovician), McElhinny and Opdyke, 1973; 3 = Juniata Formation (Ordovician), Collinson and Runcorn, 1960; 4 = Bloomsburg Formation (Silurian), Roy and others, 1967; 5 = Silurian sediments combined, Hicken and others, 1972, p. 20; 6 = Perry Formation (Devonian) combined, Hicken and others, 1972, p. 22; 7 = Mauch Chunk Formation (Upper Mississippian), Knowles and Opdyke, 1968.*

DISCUSSION

Of the four sites which give significantly grouped paleomagnetic vectors after demagnetization, sites 2, 3, and 5 are all from the northern half of the nepheline syenite body; site 5 is the ouachitite breccia plug, farther to the north. There is no obvious major petrologic or mineralogic feature which distinguishes these samples from those collected in the southern part of the body.

With regard to magnetic properties in general, site vector consistency correlates generally with low susceptibility and high Q ratio (especially $Q' = (J_{300}/X) \times 100$; Table 1). The amounts of magnetite and pyrrhotite both present initially and (or) now remaining vary considerably; secondary hematite (of low intrinsic susceptibility), however, appears to be present in all surface samples. We correlate the stable component of remanence with the ubiquitous supergene hematite, and instability with the varying magnetite (and pyrrhotite) content. On this basis, we judge the proportion of relict magnetite and pyrrhotite to hematite to be much lower in the four northern sites.

The orientation of the stable remanent vector ($D = 163°$, $I = 22°$), however, is such that if it is associated with secondary hematite, the latter cannot be the result of recent supergene weathering. As the Beemerville outcrop is very close to the contact of the Martinsburg Shale with the overlying Shawangunk Conglomerate, we infer that the present exposure must have been very close to the pre-Shawangunk land surface. Thus, we have concluded that the secondary hematite is largely the product of pre-Shawangunk weathering. If this interpretation is correct, the maximum age of the stable magnetization is between 435 m.y. (the radiometric age of the complex) and Late Silurian (the age of the Shawangunk Conglomerate; Kummel, 1940). The calculated virtual geomagnetic pole is generally similar to that obtained in other paleomagnetic studies of Paleozoic rocks from North America (Table 2). However, the relatively large confidence cone, together with uncertainties as to the precise age of the magnetization, detracts from the value of these results.

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